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Thermodynamic assessment of the Yb–Si system

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Abstract

The Yb–Si binary system was assessed using the CALPHAD method based on experimental data from the literature and a few new tensimetric measurements of the Yb activity in the liquid phase obtained by the Knudsen Effusion-Mass Spectrometry technique. A consistent thermodynamic description of the Yb–Si phases is presented: six intermediate phases (Yb_5Si_3, Yb_5Si_4, YbSi, Yb_8Si_{11}, Yb_3Si_5 and $YbSi_{1.74}$), three solid solution phases (Yb-rich bcc and fcc, Si-rich diamond) and the liquid solution phase were considered in the assessment. The intermediate phases were treated as stoichiometric compounds, the solid solutions as ideal and the liquid solution phase by the Redlich–Kister formalism. The calculated phase diagram and thermodynamic properties are in good agreement with available experimental data.
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Keywords: Phase diagram optimization; Yb–Si; Metal silicides; Ytterbium compounds; Thermodynamic assessment

1. Introduction

Metal silicides constitute an extremely interesting class of materials from a basic research point of view and for technological applications. Several transition metal–Si compounds are stable in drastic conditions of temperature and pressure as well as in oxidizing conditions; some of these are semiconductors while others behave as semimetals or metals. Due to their versatility, silicides are widely used as structural materials for high temperature applications, as functional constituents in microelectronic technologies and in many other advanced technology devices [1–5].

Rare earth metal (REM) silicides can be considered a subgroup of the larger class of metal silicides. The special contribution of 4f electrons to the REM–Si chemical bond gives these materials special optical and electronic properties that make them promising for applications as functional materials in optoelectronic device and in sensors [6–9]. Further interests for Yb–Si compounds arise from the valence fluctuations of the ytterbium atoms in the alloys [10–13].

Until recently the knowledge of the Yb–Si binary system was rather scarce, even concerning the number and identity of the intermediate compounds [14]. In the last few years a number of publications appeared in the literature about phase relations [15], crystallochemistry [16] and thermodynamics [17,18] of this system. Two new intermediate compounds have been identified (Yb_5Si_4, Yb_3Si_4) and a new version of the phase diagram has been drawn [16]. The complex phase relations in the Si-rich end part of the phase diagram have been accurately investigated [15] and the thermodynamic properties of formation of the intermediate phases have been determined by tensimetric measurements [18]. In a recent paper, Kubata et al. [19] investigated the crystal structure of the compound identified by Palenzona et al. [16] as Yb_3Si_4, obtaining a new accurate superstructure by X-ray diffraction and high resolution transmission electron microscopy and assessing the correct stoichiometry of this compound as Yb_8Si_{11}.

In this work the thermodynamic assessment of the Yb–Si binary system has been carried out by means of the CALPHAD method [20]. The optimization procedure is based on the experimental data reported in the literature [15–18].

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2. Experimental background

As discussed in the introduction, the basic knowledge about the Yb–Si binary system was rather scarce until recently regarding the identity of the intermediate phases, and completely missing as regards the phase diagram and the thermodynamic properties. In the Landolt–Bornstein compilation [14] the Yb–Si system is reported to include four intermediate phases with known crystal structure (Yb_5Si_3 hP16 Mn_5Si_3-type, YbSi oC8 CrB-type, Yb_3Si_5 hP6 Th_3Pd_5-type and $YbSi_{1.74}$ hP3-y AlB_{2-x}-type) and no phase diagram is reported, but a tentative one drawn in analogy with the Eu–Si system is sketched.

In the recent literature two experimental descriptions of the phase relations in the Yb–Si binary system have been reported by Grytsiv et al. [15] and by Palenzona et al. [16]. In these two papers a systematic investigation of the phase diagram of the silicon-rich part (Si > 62.5 at.%) [15] and of the almost complete composition range (0–70 at.% Si) [16], respectively. In particular Grytsiv et al. investigated the complex crystallochemistry of the $YbSi_{1.74}$ and Yb_3Si_5 phases and described the composition–temperature stability regions in the Si-rich part of the phase diagram. These authors reported the incongruent melting temperature of the $YbSi_{1.74}$ compound, the peritectoid and eutectoid decomposition temperatures of Yb_3Si_5 and $YbSi_{1.74}$ and also an estimation of the very small solubility range (<1 at.% Si) for the compound $YbSi_{1.74}$.

Palenzona et al. [16] investigated the phase diagram in a larger composition range (0–70 at.% Si). Two new intermediate compounds were found, namely the Yb_5Si_4 oP36 Sm_5Ge_4-type and Yb_3Si_4 oC18.67 Ho_3Si_4-type compounds. The crystallographic structure of the latter was subsequently investigated by Kubata et al. [19] who refined a new accurate Yb_8Si_{11}-type oI38 cell and redetermined the nominal stoichiometry. It is to be noted that Kubata et al. also discussed the occurrence of further limited vacancies in the Yb sublattice that could shift the stoichiometry of the compound to $YbSi_{1.4}$ instead of $YbSi_{1.375}$ (Yb_8Si_{11}). However, as the Yb defect is apparently very limited, in the optimization this phase is treated as a line-compound with stoichiometry Yb_8Si_{11}. In Table 1 a summary of the crystal structure and lattice parameters for the six known intermediate phases of the Yb–Si system is reported. It is to be noted that also the $YbSi_{1.74}$ phase is a defect compound: however, it appears to be more similar to a line-compound than to a solid solution compound. Indeed its stoichiometry, as assessed by Palenzona et al. [16], is very well defined and any homogeneity range is expected to be very narrow as verified by Grytsiv et al. [15].

In their study Palenzona et al. measured the invariant congruent melting temperature for YbSi, the incongruent melting temperatures for Yb_5Si_3, Yb_5Si_4 and $YbSi_{1.74}$, the *liquidus* curves, the peritectoid decomposition temperatures of Yb_3Si_4 and Yb_3Si_5 and the eutectoid decomposition temperature for $YbSi_{1.74}$.

Table 1
Crystal structure of the intermediate phases of the Yb–Si system

Phase	Pearson symbol	Prototype	*a* (nm)	*b* (nm)	*c* (nm)
Yb_5Si_3	hP16	Mn_5Si_3	0.8264(1)		0.6254(1)
Yb_5Si_4	oP36	Sm_5Ge_4	0.7262(2)	1.4784(4)	0.7700(2)
YbSi	oC8	CrB	0.4180(1)	1.0314(2)	0.3762(1)
Yb_8Si_{11}	oI38	Yb_8Si_{11}	0.416	0.756	2.351
Yb_3Si_5	hP8	Th_3Pd_5	0.6515(1)		0.4094(1)
$YbSi_{1.74}$	hP3	AlB_2	0.3781(1)		0.4101(1)

Data are from Ref. [16,19]; standard deviations are reported in brackets.

A summary of the experimental invariant temperatures of the Yb–Si compounds retrieved from literature and used in the optimization is reported in Table 2.

The only investigation of the thermodynamic properties of formation of the intermediate phases was reported by Brutti et al. [18]. These authors investigated by means of the Knudsen Effusion-Mass Spectrometry (KEMS) technique the high temperature decompositions of the intermediate compounds and derived the enthalpies of formation from the relevant reaction enthalpies reduced to 298 K. The general equation of the high temperature decompositions is the following:

$$\frac{z}{zx-wy}\mathrm{Yb}_x\mathrm{Si}_y(\mathrm{s}) = \mathrm{Yb}(\mathrm{g}) + \frac{y}{zx-wy}\mathrm{Yb}_w\mathrm{Si}_z(\mathrm{s}).$$

In the present work the primary tensimetric data reported in [18] were used directly in the assessment as linear combinations of the Gibbs energies of formation of the relevant ytterbium silicides for each experimental measurement. Indeed from the measured pressure of gaseous monatomic ytterbium, $p_{\mathrm{Yb(g)}}$, the following general relation is easily derived:

$$\frac{y}{zx-wy}\Delta_f G_T^o(\mathrm{Yb}_w\mathrm{Si}_z(\mathrm{s})) - \frac{z}{zx-wy} \times \Delta_f G_T^o(\mathrm{Yb}_x\mathrm{Si}_y(\mathrm{s})) = -RT\cdot\ln(p_{\mathrm{Yb(g)}}/p^o_{\mathrm{Yb(g)}})$$

in which $\Delta_f G_T^o(i)$ is the Gibbs energy of formation at temperature T of the compound i and R is the gas constant. It is to be emphasized that each vapor pressure data was used directly as input in the optimization.[1] It is also to be noted that in view of the newly assessed composition (Yb_8Si_{11}) for the compound previously indicated as Yb_3Si_4, the relevant vapor pressure data have been referred to decomposition reactions involving the new stoichiometry instead of that (Yb_3Si_4) used by Brutti et al. [18]. The enthalpies of formation and the eutectoid and peritectoid decomposition temperatures of Yb_8Si_{11}, Yb_3Si_5 and $YbSi_{1.74}$, derived from the vapor pressure data and reported in [18], were only used for a posteriori comparison with the assessed results. Indeed including these derived data in the optimization

[1] Owing to the large number of data points (157), the vapor pressure data listing is omitted in present paper. The complete set of optimization input data is available from the corresponding author.

Table 2
Incongruent and congruent melting temperatures, liquidus temperatures and peritectoid and eutectoid decomposition temperatures reported in the experimental investigations of the phase diagram [15,16]

Equilibrium	at.% Si	T (K)	Ref.
Liquidus + Yb_5Si_3	5	1443	[16]
	10	1612	
	15	1693	
	20	1785	
	25	1839	
	30	1873	
	35	1891	
Liquidus + Yb_5Si_4	40	1936	
Liquidus + YbSi	44.4	1965	
	45	1925	
	57.1	1875	
	60	1830	
	62.5	1727	
Liquid + YbSi = $YbSi_{1.74}$	65	1698	
Liquidus + $YbSi_{1.74}$	70	1661	
Liquid = YbSi	50	1983	
Yb_5Si_3 = Liquid + Yb_5Si_4	40	1903	
Yb_5Si_3 = Liquid + Yb_5Si_4	44.4	1903	
Yb_5Si_4 = Liquid + YbSi	44.4	1953	
$YbSi_{1.74}$ = Liquid + YbSi	55	1698	
	57.1	1700	
	60	1698	
	62.5	1698	
Liquid = $YbSi_{1.74}$ + Si	65	1421	
	70	1418	
	80	1418	
Yb_8Si_{11} = YbSi + $YbSi_{1.74}$	55	1243	
	57.1	1253	
	60	1253	
$YbSi_{1.74}$ = Yb_3Si_5 + Si	65	1038	
	70	1043	
	80	1043	
Yb_3Si_5 = Yb_8Si_{11} + $YbSi_{1.74}$	62.5	1238	[16]
Liquid = Yb(fcc) + Yb_5Si_3	5	1088	
	15	1086	
	20	1088	
	30	1085	
	35	1087	
Yb(bcc) = Yb(fcc)	5	1053	
	10	1053	
	20	1053	
	25	1053	
	30	1053	
	35	1053	
Liquidus + YbSi	61	1733	[15]
	62.5	1743	
	64	1733	
	66	1708	
Liquidus + $YbSi_{1.74}$	70	1643	
Liquidus + Si	87	1543	
$YbSi_{1.74}$ = Liquid + YbSi	61	1688	
	62.5	1676	
	65	1683	
	66	1673	
Liquid = $YbSi_{1.74}$ + Si	64	1408	
	66	1408	
	70	1408	
	87	1401	
$YbSi_{1.74}$ = Yb_3Si_5 + Si	64	1038	
	66	1033	
	70	1038	
	87	1023	

procedure would be redundant. From vapor pressure data reported in [18] it is possible to derive a complete set of standard entropies of formation entropies at 298 K following the same second-law method used by the authors to obtain the enthalpies of formation: also, these determinations have been used only for comparison because they are directly derived from the vapor pressure data. It is to be noted that while enthalpies and entropies of formation usually show the same sign, in the present case enthalpies are negative whereas entropies are positive.

The only other available experimental data are the low temperature (0–60 K) heat capacities for the Yb_5Si_4, YbSi and Yb_3Si_5 phases reported by Dhar et al. [18]. Obviously these determinations cannot be used directly in the optimization as SGTE data [21] are referred to 298 K and are valid only above this temperature. Moreover, as these data do not allow for a complete integration in the 0–298 K temperature range of C_p^o/T, approximate extrapolations of the heat capacities to 298 K by means of the Nernst–Lindermann and Tarassov equations [22,23] have been attempted in order to obtain the entropies of formation of the three investigated compounds. The so-derived C_p^o have been used to estimate the formation entropies of the Yb_5Si_4, YbSi and Yb_3Si_5 compounds at 298 K. Because these values are rough estimates, they have been used only for comparison with the assessed data.

All the thermodynamic data taken or derived from Ref. [18] are discussed in Section 3.

Besides the literature data some new Yb activity data have been measured studying the high temperature reaction $YbSi_{1.74}(s)$ = Yb(g) + liquid solution. These data have been obtained by the Knudsen Effusion-Mass Spectrometry technique, following the same experimental procedure described in Ref. [18]. Tensimetric experiments have been carried out using a 60° magnetic sector, single focusing mass spectrometer (Nuclide HT-7), coupled with a Knudsen molecular source. Molybdenum Knudsen cells have been used. Temperature was measured using a Pt/Pt–Rh (10%) thermocouple in direct contact with the bottom of the cell. More details about the experimental apparatus and data analysis can be found elsewhere [18,24]. Vapor pressure measurements have been performed starting from a 65 at.% Si sample ($YbSi_{1.74}$–Si) heated above the Si-rich eutectic reaction temperature (1421 K). The composition before and after the experiments has been checked by X-ray diffraction and scanning electron microscopy electroprobe

microanalysis. Ytterbium activities have been derived by comparing the measured Yb vapor pressures with tabulation derived data [21] for pure ytterbium. Data are shown in Table 3.

Table 3
Yb activities measured in the $YbSi_{1.74}$-liquid solution two-phase region (at.% Si ∼ 65%)

T (K)	Yb activity	T (K)	Yb activity
1432	1.1×10^{-05}	1434	1.2×10^{-05}
1421	1.4×10^{-05}	1465	2.0×10^{-05}
1450	1.2×10^{-05}	1500	3.2×10^{-05}
1485	1.3×10^{-05}	1512	3.5×10^{-05}
1493	1.5×10^{-05}	1484	2.3×10^{-05}
1456	8.5×10^{-05}	1449	1.4×10^{-05}
1424	5.2×10^{-05}		

Ytterbium reference state is Yb(liquid).

In summary, the experimental data used in the present work are the following:

- Incongruent and congruent melting temperatures, liquidus and peritectoid and eutectoid decomposition temperatures derived from the experimental investigations of the phase diagram [15,16].
- Vapour pressure data from the high temperature decomposition reactions of the six intermediate phases, considered as linear combination between the Gibbs energy of formation of the relevant compounds, in seven different two-phase regions [18].
- Yb activity measurements, derived from vapor pressure data, in equilibrium over the liquid and $YbSi_{1.74}$ phases.

The whole set of input data used in the optimization procedure consists of 232 experimental determinations (62 phase diagram points and invariant temperatures, 157 vapour pressure data and 13 activity data in the liquid). It is to be noted that approximately 67% of the overall input dataset consisted in experimental vapor pressure derived data and the optimization output could suffer from biasing. However, it has been preferred to keep the entire input dataset in order to fully represent the available experimental data which are originated by a rather limited number of different sources.

3. Thermodynamic modeling

The assessment was performed following the CALPHAD approach [20]. In this method the stability of each phase is expressed by its molar Gibbs energy function.

The Yb–Si system includes six intermediate phases (Yb_5Si_3, Yb_5Si_4, YbSi, Yb_8Si_{11}, Yb_3Si_5 and $YbSi_{1.74}$) and four solution phases (liquid, fcc, bcc and diamond). The intermediate phases have been modeled as line-compounds with fixed stoichiometry. The molar Gibbs energy function for a general stoichiometry Yb_xSi_y was expressed by the following equation:

$$G_m^{Yb_xSi_y} = \left(\frac{x}{x+y}\right) \cdot {}^oG_{Yb}^{fcc} + \left(\frac{y}{x+y}\right) \cdot {}^oG_{Si}^{diamond} + \left(A^{Yb_xSi_y} + B^{Yb_xSi_y} \cdot T\right)$$

in which ${}^oG_{Yb}^{fcc}$ and ${}^oG_{Si}^{diamond}$ are the molar Gibbs energies for the elements in their reference state (fcc for ytterbium and diamond for silicon), and $A^{Yb_xSi_y}$ and $B^{Yb_xSi_y}$ are the parameters to be optimized, corresponding, respectively, to the formation enthalpy and negative of the formation entropy at 298 K (reference temperature) for the compound.

The molar Gibbs energy of a solution phase is expressed by the sum of three independent terms:

$$G_m^{phase} = G_m^o + G_m^{id} + G_m^{xs}$$

in which G_m^o is the contribution from the pure elements, G_m^{id} is the ideal mixing term and G_m^{xs} is the excess term. By expressing the excess term by the Redlich–Kister formula the previous equation becomes:

$$G_m^{phase} = \left(\chi_{Yb} G_{Yb}^{phase} + \chi_{Si} G_{Si}^{phase}\right) + RT(\chi_{Yb} \ln \chi_{Yb} + \chi_{Si} \ln \chi_{Si}) + (\chi_{Si}\chi_{Yb}\Omega_{Si-Yb})$$

in which G_{Yb}^{phase} and G_{Si}^{phase} are the Gibbs energies of the pure elements in the solution phase, χ_{Yb} and χ_{Si} are the mole fraction of Yb and Si and Ω_{Si-Yb} is the interaction parameter. This parameter can be a function of composition and temperature. Following the Redlich–Kister formalism its explicit dependence by the composition is given by:

$$G_m^{xs} = (\chi_{Si}\chi_{Yb}) \sum_{\nu} ({}^{\nu}\Omega_{Si-Yb} \cdot (\chi_{Si} - \chi_{Yb})^{\nu})$$

in which ${}^{\nu}\Omega_{Si-Yb}$ is the interaction parameter of n-order with respect to the solution composition. The temperature dependence of ${}^{\nu}\Omega_{Si-Yb}$ is usually expressed in the linear form:

$${}^{\nu}\Omega_{Si-Yb} = A_{\nu} + B_{\nu} \cdot T.$$

In the Yb–Si system the homogeneity range of the terminal solid solutions bcc, fcc and diamond cubic are very narrow: the solid solubilities of Yb in the Si-diamond lattice and Si in the Yb-(fcc or bcc) lattices are less than 1% [16]. Due to the absence of experimental or computational determinations of the Si–Yb interaction parameters in these lattices the three solid solutions (bcc, fcc and diamond) have been treated as ideal.

On the other hand the Yb–Si liquid phase is expected to deviate significantly from ideality as several Me–Si systems do [25–27]. For example Grobner et al. [25] in their assessment of the calcium–silicon system treated the liquid phase as a sub-regular solution. As ytterbium usually shows a rather similar chemical behavior compared to calcium, a similar thermodynamic modeling has been adopted for the Yb–Si liquid by expressing the excess contribution in the

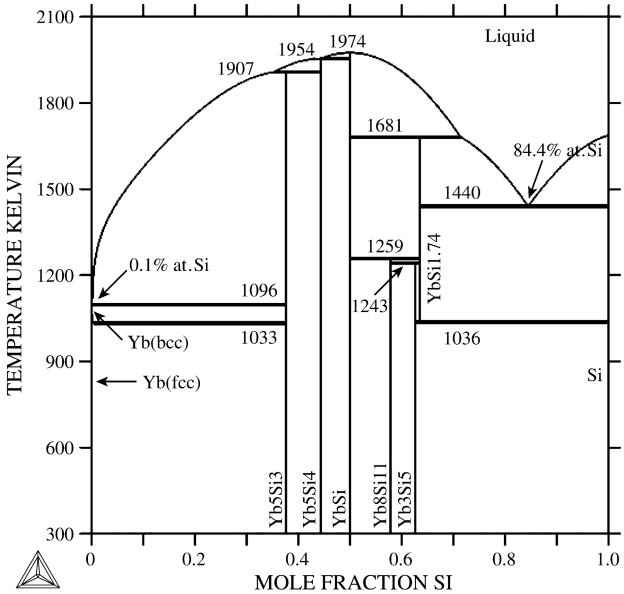

Fig. 1. Assessed phase diagram of the Yb–Si system.

following form:

$$G_m^{xs} = (\chi_{Si}\chi_{Yb})[(A_0 + B_0 \cdot T) + (A_1) \cdot (\chi_{Si} - \chi_{Yb})].$$

It is to be noted that in the case of Yb–Si, compared to the Ca–Si liquid phase, only three interaction parameters were necessary to achieve for a satisfactory thermodynamic modeling. The introduction of a fourth interaction parameter for the Yb–Si liquid phase has been tested and it resulted in a improved overall accuracy of the assessment. However, the thermodynamic function of mixing of the liquid showed unusual shapes, leading to an unreliable overall thermodynamic description.

In summary the thermodynamic modeling of the Yb–Si system includes 15 independent parameters to be optimized, 12 for the intermediate line-compounds and 3 for the liquid phase.

4. Optimization results

The evaluation of the thermodynamic parameters has been carried out using the PARROT [28] module of the Thermo-Calc software package [29]. Data for the pure elements have been taken from the SGTE compilation [21]. The assessed parameters are reported in Table 4 and the calculated phase diagram is shown in Fig. 1. In Fig. 2 the calculated diagram is compared with the experimental data points from [15,16,18]. The phase diagram calculated by the assessed parameters reported in Table 4 is in good agreement with the experimental data, with only the liquidus curve in the composition range 55–65 at.% Si slightly deviating somewhat significantly. In Table 5 a summary of the invariant equilibria of the binary Yb–Si system is reported in order to compare with the experimental and assessed temperatures: satisfactory agreement is achieved

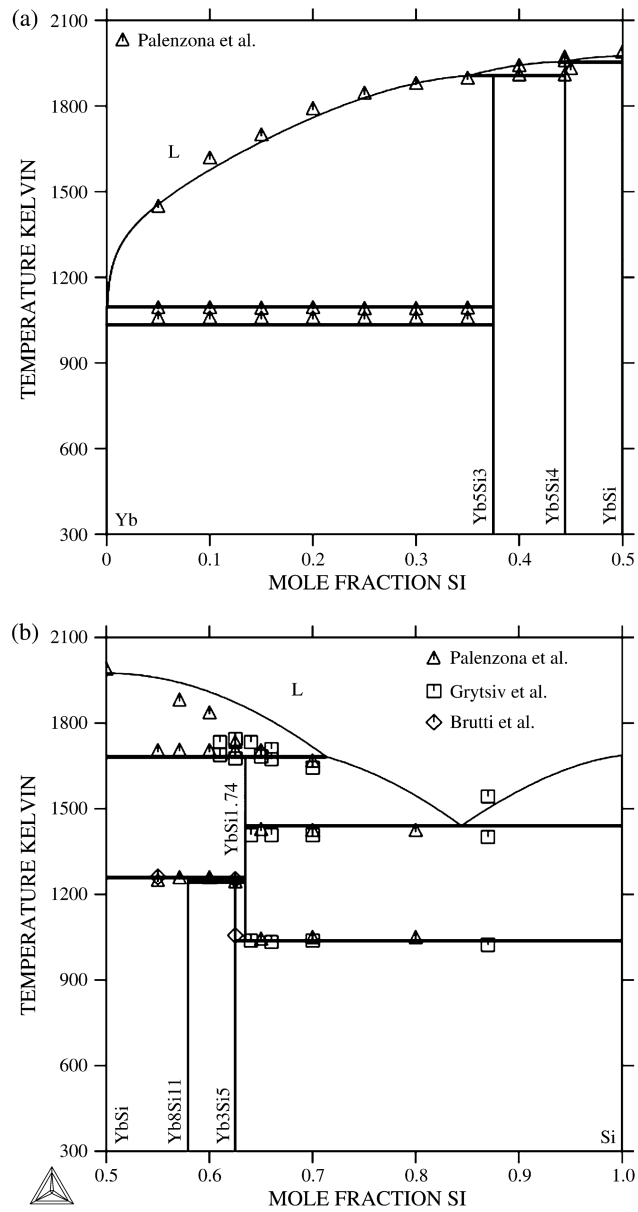

Fig. 2. Comparison between the optimized diagram and the experimental data points taken from [15,16,18]: (a) Yb-rich, (b) Si-rich.

except for the Si-rich eutectic temperature which deviates as much as 22 K from the experimental data [16]. In particular the agreement between experimental and computed high temperature congruent and incongruent melting points of Yb_5Si_3, Yb_5Si_4, YbSi and $YbSi_{1.74}$ is remarkable. It is also worth noting the agreement obtained with the invariant temperatures reported by Brutti et al. [18] that were not included in the optimization input dataset. In passing, the poor agreement between experimental and calculated transition temperature for the Yb-rich fcc–bcc solid solution transformation should be mentioned. Experimentally this transition for the pure Yb takes place at 1005 K [30] and at 1053 K for the Yb-rich Yb–Si alloys [16], while the computed temperature is 1033 K. However, the shift in the

Table 4
Optimized parameters from the assessment of the thermodynamics of the Yb–Si system (energies are expressed in J mol atoms^{-1})

Liquid phase	
$G_m^{\text{liquid}} = \left(\chi_{\text{Yb}} G_{\text{Yb}}^{\text{liquid}} + \chi_{\text{Si}} G_{\text{Si}}^{\text{liquid}}\right) + RT \cdot (\chi_{\text{Yb}} \ln \chi_{\text{Yb}} + \chi_{\text{Si}} \ln \chi_{\text{Si}})$ $+(\chi_{\text{Si}}\chi_{\text{Yb}}) \cdot [(A_0 + B_0 \cdot T) + (A_1) \cdot (\chi_{\text{Yb}} - \chi_{\text{Si}})]$	
A_0	−51 716.2
B_0	−67.1012
A_1	−29 995.8

Intermediate phases		
$G_m^{\text{Yb}_x\text{Si}_y} = \left(\frac{x}{x+y}\right) \cdot {}^oG_{\text{Yb}}^{\text{fcc}} + \left(\frac{y}{x+y}\right) \cdot {}^oG_{\text{Si}}^{\text{diamond}} + \left(A^{\text{Yb}_x\text{Si}_y} + B^{\text{Yb}_x\text{Si}_y} \cdot T\right)$		
Compound	A	B
Yb_5Si_3	−49 792.5	−4.22282
Yb_5Si_4	−51 091.3	−6.19451
YbSi	−50 974.2	−7.38652
Yb_8Si_{11}	−49 106.9	−3.88883
Yb_3Si_5	−45 026.2	−4.20409
$YbSi_{1.74}$	−41 125.3	−6.69761

Table 5
Invariant equilibria of the binary system Yb–Si (L = liquid phase, m.t. = melting temperature)

Invariant reaction	Type	Temperature (K)			
		This work	[16]	[15]	[18]
Yb(bcc) = Yb(fcc)	Degenerate	1033	1053[a]		
L = Yb(bcc) + Yb_5Si_3	Eutectic	1096 0.1 at.% Si	1088 <1 at.% Si		
L + Yb_5Si_4 = Yb_5Si_3	Incongruent m.t.	1907	1903		
L + Yb_5Si_4 = YbSi	Incongruent m.t.	1954	1953		
L = YbSi	Congruent m.t.	1974	1983		
L + YbSi = $YbSi_{1.74}$	Incongruent m.t.	1681	1698	1681	
L = Si + $YbSi_{1.74}$	Eutectic	1440 84.4 at.% Si	1418 81 at.% Si	1408 80.6 at.% Si	
$YbSi_{1.74}$ + YbSi = Yb_8Si_{11}	Peritectoid	1259	1253		1261
Yb_8Si_{11} + $YbSi_{1.74}$ = Yb_3Si_5	Peritectoid	1243	1238		1252
$YbSi_{1.74}$ = Yb_3Si_5 + Si	Eutectoid	1036	1043	1036	1056

[a] This transition temperature refers to Palenzona et al. measurements performed on Yb-rich Yb–Si samples [16]. The assessed data for pure ytterbium reported in literature [30] is $T_{\text{Yb(bcc)=Yb(fcc)}} = 1005$ K.

computed fcc–bcc transition is only apparent because the modeling gives a value that is identical compared to the computed transition temperature for the pure Yb transition.

In Table 6 is shown a comparison between the enthalpies of formation $\Delta_f H_{298}^o$ (kJ/mol atoms) and the standard entropies of formation $\Delta_f S_{298}^o$ (J/K mol atoms) for the intermediate phases derived from the present optimization or from literature data [17,18]. As discussed in the previous section, these results were not included in the input dataset, but used as a check of the assessed results. Also, in this case the agreement is satisfactory for both entropies and enthalpies of formation. It is to be noted that while literature data shown in Table 6 were not included in the optimization these are directly related to part of the input dataset used (see previous section), except for the formation entropies derived by Dhar et al. from low temperature heat capacities [17] that are completely independent. On the other hand, the agreement achieved is still notable because it underlines the consistency of the experimental determinations used. Indeed the optimization has been derived from a set of data that included not only vapor pressure measurements but also phase diagram determinations. The agreement obtained with the only partially related thermodynamic data confirms the self-consistency and reliability of the present assessment.

As already mentioned, the entropies and the enthalpies of formation of all the ytterbium silicides are positive and negative respectively, while usually these show the same sign for intermetallic compounds. It is to be noted that this

Table 6
Enthalpies of formation $\Delta_f H^o_{298}$ (kJ/mol atoms) and standard entropies of formation $\Delta_f S^o_{298}$ (J/K mol atoms) for the intermediate phases of the Yb–Si, derived from the present optimization and taken from literature

	$\Delta_f H^o_{298}$		$\Delta_f S^o_{298}$		
	This work	[18]	This work	[18]	[17]
Yb_5Si_3	−49.8	−48.3 ± 3.6	4.2	4.5	
Yb_5Si_4	−51.1	−53.2 ± 4.6	6.2	2.8	1.3
YbSi	−51.0	−51.1 ± 5.1	7.4	4.9	0.8
Yb_8Si_{11}	−49.1	−48.0 ± 3.1	3.9	3.3	
Yb_3Si_5	−45.0	−41.3 ± 2.6	4.2	5.2	0.4
$YbSi_{1.74}$	−41.1	−37.4 ± 0.9	6.7	7.6	

unusual behavior is similar to that of calcium silicides as reported in [25].

In Fig. 3 the computed thermodynamic functions of mixing (Gibbs energy, enthalpy and entropy) at 2000 K for the liquid phase are shown. Although no comparison with literature is possible due to the lack of experimental determinations, some comments are appropriate. The thermodynamic functions of mixing of the liquid phase have been modeled by the liquidus data and the few activity data reported in Table 3; therefore the extrapolation at 2000 K shown in Fig. 3 should be considered as estimates consistent with the overall modeling. Experimental investigation of thermochemical properties of the Yb–Si liquid phase at high temperature in the whole composition range would be useful in order to validate these calculated values.

In summary, the thermodynamic assessment of the binary system Yb–Si has been carried out by means of the CALPHAD approach using the Thermo-Calc software. An input dataset has been used including phase diagram determinations and vapor pressure data. Some new Yb activity measurements performed in the two-phase region $YbSi_{1.74}$-liquid by means of the KEMS technique have been reported and considered in the assessment. The optimization output compares well with the input data and also with a further validation dataset not directly included in the assessment procedure.

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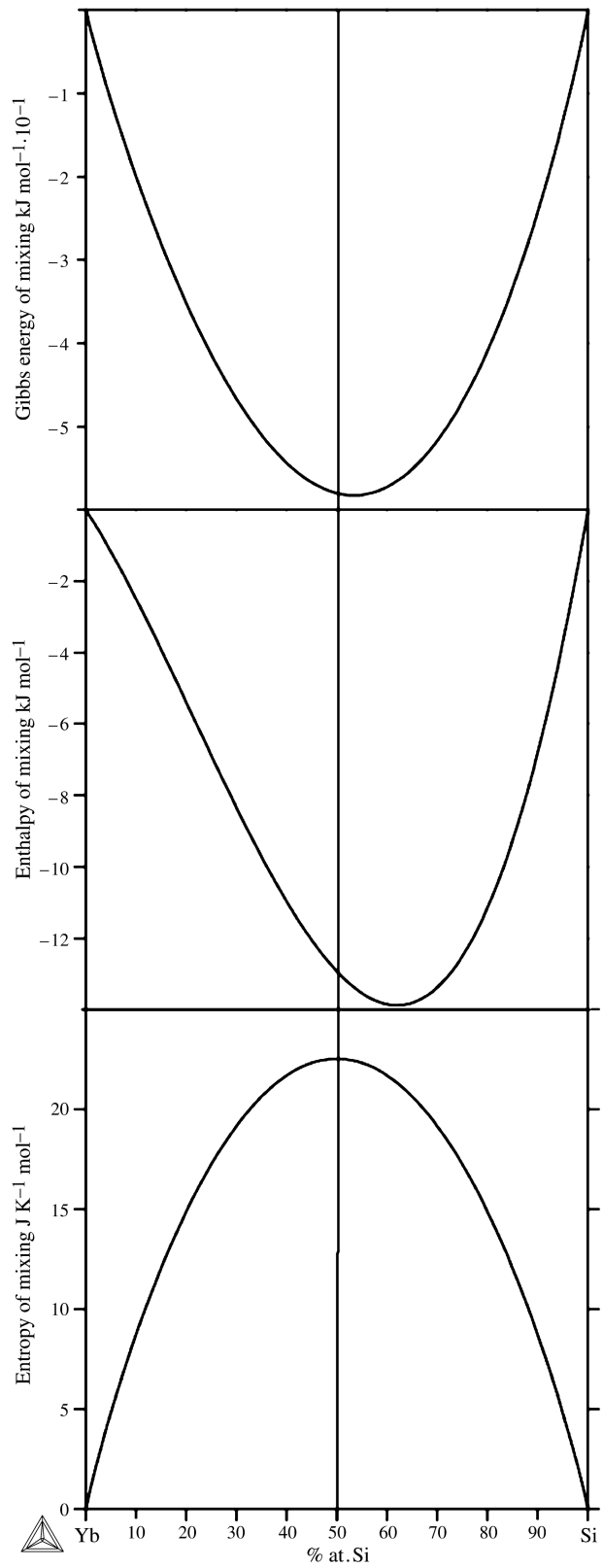

Fig. 3. Thermodynamic function of mixing of the liquid phase at 2000 K. Reference states for Yb and Si are the relevant pure liquid phases.

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